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Computer Vision and Voice Assisted Image Captioning Framework for Visually Impaired Individuals using Deep Learning Approach

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***Abstract-*. Blind or visually challenged have considerable obstacles when obtaining visual material, restricting their capacity to derive meaningful information from pictures. This research proposing a novel framework combining computer vision with voice-based image captioning using deep learning techniques. The proposed system employs VGGNet-16 model for photo processing and long short-term memory networks (LSTMs) for natural language processing. The models used in this research were trained on Flickr8k, Flickr30k, and a bespoke dataset. Subsequently, these models were deployed on a Raspberry Pi 4B single-board computer that was equipped with a graphics processing unit. A two-fold approach is used, whereby the first phase entails exposing the input image to preprocessing using a pre-established VGGNet-16 model. This methodology enables the retrieval of relevant visual attributes, therefore capturing the intrinsic semantic content of the picture. The aforementioned attributes are fed into a language model based on LSTM to generate explanatory captions. To enhance communication with visually impaired people, the generated captions are converted into audible speech using text-to-speech synthesis technology. The effectiveness of the proposed architecture is evaluated via extensive experiments using benchmark datasets and real-time images obtained with a NoIR camera unit. The generated captions are evaluated using quantitative assessment criteria, namely BLEU and ROUGE scores. The developed VGGNet16 model has exceptional performance in terms of accuracy (95.620%), precision (96.928%), recall (87.879%), and F1 score (92.182%). The results suggest that using computer vision in conjunction with voice-based image captioning offers a promising solution for addressing the challenges faced by those with visual impairments in perceiving and comprehending visual content.**

Keywords- computer vision, image captioning, LSTM, self-attention, transformer encoder, visually impaired, VGGNet-16.

I. INTRODUCTION

The worldwide landscape of supporting the visually impaired has changed due to the confluence of technological advancement and societal shifts, creating an environment that is more welcoming and accommodating to those with visual impairments. The prevalence of wearable devices such as smart glasses has increased, including enhanced sensors, cameras, and artificial intelligence-driven algorithms [1,2]. Individuals who are blind or have visual impairments sometimes have difficulties accessing visual information in their surroundings, including textual materials, images, and photos. The use of voice-assisted technology for picture captioning and navigation provides people with a means to get information about the visual elements of pictures and situations, hence enabling a deeper understanding of their environment [3]. By receiving accurate and detailed auditory descriptions of images and scenes, individuals can make informed decisions, navigate unfamiliar surroundings, and engage with visual information independently, without assistance from sighted individuals.

The integration of computer vision (CV), speech recognition, artificial intelligence (AI), and deep learning (DL) and Internet of things (IoT) techniques has resulted in the emergence of a novel framework known as the computer vision-aided voice-based picture captioning framework [4,5]. The suggested framework had the potential to revolutionize significantly the lives of those who experience blindness or visual impairment since it offers them unequalled access to visual information. A graphics processing unit (GPU) derived from the Raspberry Pi 4B is utilized to develop and implement the model. The Raspberry Pi 4B single board computers (SBC) have been purposefully designed to efficiently carry out several tasks simultaneously, enabling real-time processing of images, analysis of images, and natural language processing. This research investigates the capabilities of a No Infrared (NoIR) camera to capture images in low light conditions, hence enhancing nighttime vision technology. This capability can potentially improve the safety and mobility of those who are blind or visually impaired by facilitating their navigation in low-light conditions.

The new approach of incorporating the long short-term memory (LSTM) and visual geometric group (VGGNet-16) models into voice-assisted picture captioning for those with visual impairments leverages the strengths of these deep learning architectures to provide comprehensive image descriptions [4]. The VGG16 model has been pre-trained using a substantial ImageNet dataset, demonstrating its proficiency in accurately detecting and categorizing various visual patterns and objects seen in images. The LSTM is a distinct kind of recurrent neural network (RNN) architecture designed to handle sequential input efficiently. The attribute

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above renders LSTM especially well-suited for generating textual descriptions. The voice-assisted image captioning system uses an LSTM model to process the extracted image features derived from the VGG16 model. The LSTM model is responsible for creating descriptive captions. The textual descriptions are then converted into audible speech using a text-to-speech (TTS) application programming interface (API). The integrated LSTM-VGG16 model can operate in real-time, expeditiously allowing users to get visual descriptions through voice assistance. This feature enhances the individual's ability to investigate and interact with their surrounding environment independently.

Voice-assisted photo captioning is a mechanism for establishing a connection between those who do not possess conventional visual perception and the visual realm. By fulfilling these essential requirements, this technology plays a vital role in enhancing the autonomy of individuals with visual impairments, elevating their quality of life, and enabling their full participation in a visually-oriented society. In the following sections, we will undertake a more thorough analysis of the components of this methodology and the cutting-edge technology used in picture captioning. Chapter 3 delves into the examination of the technique and training procedure while also engaging in a scholarly discussion on the challenges and potential opportunities within the field of photo captioning using VGG16 and LSTM networks. Chapter 4 of this research comprehensively examines the suggested framework via experimental analysis. Following this analysis, the research concludes with a discussion of its findings and implications and potential avenues for future research in this field.

II. RELATED WORK

The task of image captioning entails the generation of written descriptions that effectively represent the content and context of an image, thereby facilitating the comprehension and communication of visual information by computers. A. K. Poddar et al. [6] present a neural network model consisting of many layers of convolutional neural networks (CNN) and LSTM units. Multiple models were trained by altering hyperparameters and the number of hidden layers to identify the optimal model and maximize the probability of the resulting Hindi captions from images. It is essential to consider the requirements of visually impaired persons across different languages within a global framework. In their study, Ganesan J al. [7] expanded upon the VGG16-LSTM framework to provide multilingual picture captioning, providing visually impaired individuals with the ability to access material in their desired language. The study conducted by R. saleem et al. [8] emphasized the significance of using the VGG16-LSTM framework in producing informative captions for pictures, which can audibly have conveyed to those with visual impairments via text-to-speech technology. This technique has shown the potential of picture captioning to foster a more inclusive digital environment. Xiao et al. [9] presented a novel captioning technique that combines two distinct LSTM networks using an adaptive semantic attention model. The initial LSTM network is accompanied by an attention model, capable of dynamically balancing the importance of visual semantic region and textual content. The second LSTM serves as a language model, integrating the hidden state representation of the first LSTM and the attention context vector and subsequently generating the sequence of words.

Recently, the use of the encoder-decoder architecture, coupled with an attention mechanism, has shown significant advancements in picture captioning. Tan et al. [10] provide a novel approach for image captioning that utilizes a hierarchical phi-LSTM architecture to create descriptive captions for images. The system is designed to convert picture captions from phrase to sentence format using a phrase decoder that decodes noun phrases of varying lengths and an abbreviated sentence decoder that converts the image description into an abbreviated version. In recent years, the existing models that rely on LSTM for generating descriptions use a sequential approach, which hinders parallel processing and fails to consider the hierarchical structure of the captions adequately. To address this issue, Bai et al. [11] utilizes a CNN-based image caption generation model using conditional generative adversarial training (CGAN) as a supportive technique. In addition, using a multi-modal graph convolution network (MGCN) enables the exploration of visual associations among objects that can effectively capture and represent visual associations among items within an image.

The present picture captioning models need a better accuracy rate in accurately describing the image's content. This inaccuracy can be attributed to erroneous descriptions or a lack of consideration for scene identification. Peng et al. [12] offer a model that aims to identify the scene information matching the text volume of the LDA analysis corpus. The use of a ResNet model facilitates the extraction of global picture features, as well as the extraction of deep scene features. During the generation process, a double Long Short-Term Memory (LSTM) model is used to optimize the parameters and enhance the precision of statement production. Lim et al. [13] introduce the Mask Captioning Network (MaC) with an object layer and a backdrop layer. The authors use the Mask RCNN framework to identify prominent areas at the pixel. Furthermore, the MaC framework has been included in image captioning systems using both LSTM-based and Transformer-based models. The use of transformers has dramatically improved the efficacy of image description models [14,15]. However, the attention mechanism in transformers needs to be improved in its ability to capture intricate correlations between key and query vectors. Ji, W et al. [16] propose a novel double-attention architecture that enhances the encoder-decoder structure for addressing picture captioning challenges. The authors propose a method that enhances the performance of picture description by improving Self-Attention (SA) from two perspectives including a Masked Self-Attention and a hybrid weight distribution (HWD) module.

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One of the challenges encountered in picture captioning is the possible bias that can exist within datasets. This bias can adversely affect the quality and pertinence of the generated captions [17]. Integrating user input is crucial in developing image captioning systems that are efficient for those with visual impairments. In summary, the VGG16 and LSTM networks for picture captioning has shown significant promise in improving accessibility for those with visual impairments. The literature review highlights the significant progress in developing inclusive technologies that use artificial intelligence to provide comprehensive and precise descriptions of visual material.

III. METHODOLOGY

The deployment of the VGG16 model on a GPU platform, specifically designed for image classification, requires the implementation of specific preprocessing processes to prepare the input images for accurate feature extraction. The NoIR camera is used to capture video frames, which are then converted into images before undergoing preprocessing. The VGG16 model requires input images to conform to a predefined size. The photographs are downsized to dimensions of 224x224x3 pixels to conform to the model's input parameters.

The photographs that had been resized underwent a normalization process to achieve a consistent scale. Nevertheless, the VGG16 model necessitates input pictures to possess three channels. As a result, grayscale pictures are replicated throughout all three color channels to fulfil this need. The implementation employs a batching strategy, whereby a collection of preprocessed photographs is grouped into a batch to enhance the processing efficiency. Once the preprocessing operations have been carried out, the images are prepared for input into the VGG16 model to facilitate the feature extraction process.

Figure 1 illustrates the use of word embedding, a method employed in LSTM networks, particularly within the domain of natural language processing tasks, such as the generation of descriptive captions for images. The process of word embedding entails the transformation of individual words or tokens into vector representations characterized by continuous values. Generating a reference involves the compilation of unique words extracted from the captions in the training dataset. The dataset used in this research comprises a total of 5,000 tokens. During the training procedure, the word embedding experiences simultaneous updates with the other parameters of the model. The image captioning model, VGG16-LSTM, combines preprocessed image features retrieved from the VGG16 model with embedded words produced from the word embedding layer.

Feature vectors are often associated with the internal representations produced by the LSTM model at each time step when processing a sequence. Long short-term memory networks are a kind of recurrent neural networks (RNNs) that have been particularly designed to handle sequential data properly. They can capture long-term relationships in the data and address the problem of disappearing gradients. The LSTM network is presented with an input vector at every time step within the input sequence. The input vector might be an encoded word or a feature vector derived from the previous layer. The input vector has the potential to be encoded by an embedded word derived from the previous time step. The memory cells undergo updates at each time step, which are decided by the current input, the previous memory state, and the candidate values provided by the gates of the LSTM model. The feature vector is derived from the hidden state at each time step. The vector represents the essential information obtained from the input sequence. Using the feature vector at the last time step facilitates the production of the ultimate caption in the context of photo captioning.

IV. EXPERIMENTAL ANALYSIS

A. Dataset

Commonly used datasets for training and evaluating photo captioning systems include publicly available collections such as Flickr8k [18], Flickr30k [19], and bespoke datasets. The dataset known as Flickr30k comprises a grand total of 31,783 photographs, whereas Flickr8k, a subset of this dataset, consists of 8,000 images that have been obtained from the widely used photo-sharing network known as Flickr. Five captions given by different annotators have accompanied each

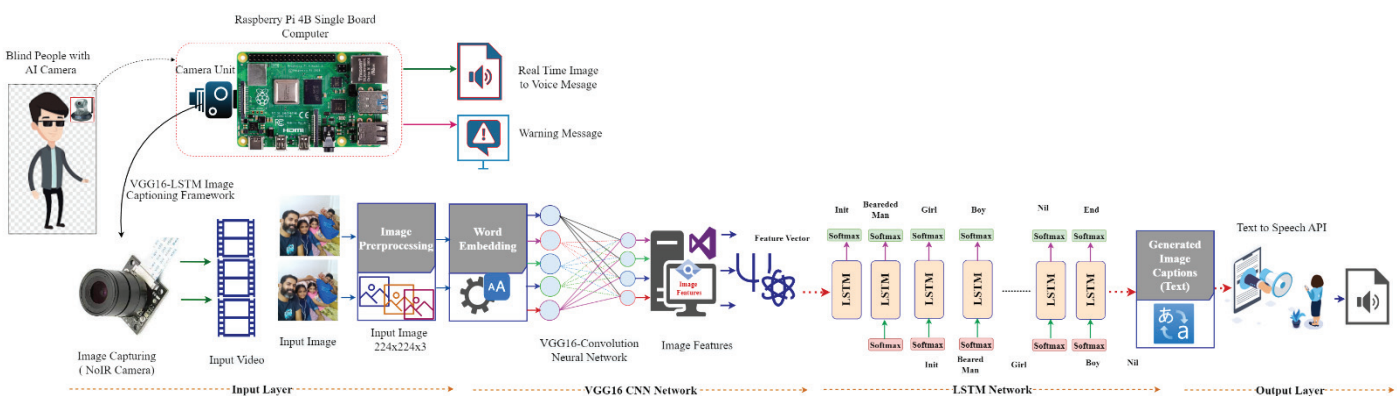

Fig. 1. A VGG16-LSTM image captioning framework developed for blind and visually impaired people.

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picture. When using these datasets, there are disparities in their dimensions, formats, and degrees of annotation accuracy. The selection of a combined dataset is conducted with meticulous consideration, considering our research aims and the complexity of our photo captioning infrastructure. Creating a customized dataset allows researchers to collect data directly relevant to their unique research question, leading to improved performance of the model. The correctness, consistency, and high quality of annotations are given priority due to their crucial role in efficiently training reliable AI models. A customized dataset has been created by considering several factors like, the techniques used for data collection, data annotation procedures, and data augmentation strategies. The training set is the largest component of the dataset, including 70% of the photographs used for model training. A validation set was created by using 15% of the dataset. A portion of 15% of the dataset is allocated for the test dataset, which is separate from the training and validation sets and is not used at any model-building step. Table 1 presents a comprehensive summary of the dataset used in this research.

TABLE I. DETAILS OF VARIOUS IMAGE DATASET USED FOR IMAGE CAPTIONING.

SI No.	Dataset name	Total No. of Training Images
1	Flickr_8k [18]	8000
2	Flickr_30k [19]	31783
3	Custom	1250
	Total	41033

From table I, we have developed the proposed model with 41k number of image with corresponding captions. Merging datasets enhances the generalization of diverse data distributions and multiple datasets let the model learn from different instances and generalize to new data. A broad training set reduces bias and overfitting, making the model more resilient and adaptable.

B. Software and Hardware

The topic of interest in this section is to pertains to the domains of software and hardware. The model above has been developed and deployed using the Raspberry Pi 4 B single-board computer. The Raspberry Pi 4 Model B is a single-board computer with a robust graphics processing unit (GPU) powered by a quad-core ARM Cortex-A72 central processing unit (CPU). Increasing memory capacity to 8 GB enables more excellent multitasking capabilities and enhanced performance, especially for programs that need significant memory resources. The board is furnished with a VideoCore VI Graphics Processing Unit (GPU) that supports OpenGL ES 3.x and facilitates the decoding of 4K videos.

Keras and TensorFlow is the major libraries used for constructing and training deep learning models specifically tailored for generating descriptive captions for images. TensorFlow serves as the fundamental framework, while Keras operates as an application programming interface (API) that provides a more abstracted interface for constructing and training neural networks. The loss function is calculated during the training process, which is a combination of the cross-entropy loss used to forecast each word inside the caption sequence. The model was constructed using TensorFlow's optimizer and loss function and then trained using a preprocessed dataset of pictures and captions. During the evaluation phase, the model that has undergone training generates descriptive descriptions for photographs. The procedure above involves the use of the trained RNN to predict individual words sequentially. The phrases that have been created are then reintroduced into the model throughout each iteration. Afterwards, adjusting the hyperparameters and model architecture becomes essential to maximize performance. This entails modifying the learning rates and dropout rates.

C. Model training and validation

In the training and evaluation phase of the proposed VGG16-LSTM image captioning system, metrics such as accuracy, sensitivity, specificity, and F1 score are used to assess the quality of the generated captions compared to the ground truth captions. Within the scope of this research, accuracy is defined as the proportion of accurately generated words in the whole caption sequence about the matching ground truth caption. The metric of sensitivity or recall assesses the model's capacity to capture the pertinent information in the ground truth captions accurately. It measures the model's ability to generate the proper words when included in the reference caption. The concept of specificity can be used to evaluate the model's capacity to avoid creating terms absent in the reference caption. The calculation involves determining the proportion of actual negative words about the overall count of irrelevant words that were erroneously generated. Another metric often used in evaluating classification models is the F1 score, the harmonic mean of accuracy and recall. The F1 score takes into account the model's ability to accurately identify pertinent terms as well as its ability to include all essential words found in the reference caption.

TABLE II. VALIDATION SCORE OF VGGNET-16 LSTM MODEL.

SI No.	Model	Accuracy (%)	Precision (%)	Recall (%)	F1 score (%)
1	VGG16	95.620	96.928	87.879	92.182

$$\text{Accuracy} = (TrP + TrN) / (TrP + TrN + FaP + FaN) \quad (1)$$

$$\text{Precision} = TrP / (TrP + FaP) \quad (2)$$

$$\text{Recall} = TrP / (TrP + FaN) \quad (3)$$

$$\text{F1 score} = 2[(\text{Precision} * \text{Recall}) / (\text{Precision} + \text{Recall})] \quad (4)$$

The evaluation of models in table II provides valuable information about their precision and efficacy. A full evaluation of the performance of a model or test in various circumstances is achieved by comprehending the concepts of true positives (TrP), true negatives (TrN), false positives (FaP), and false negatives (FaN) as discussed in equation 1 to 4.

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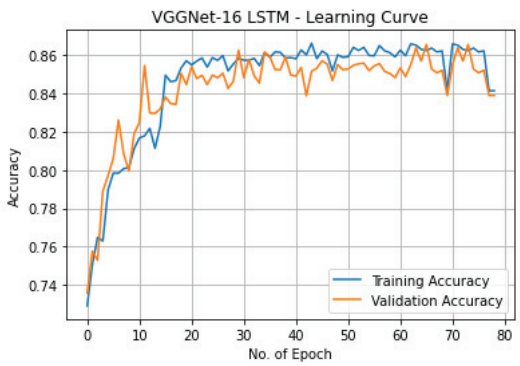


Fig. 2. Training accuracy and validation accuracy of VGGNet-16 LSTM models.

In Figure 2, the relationship between the number of epochs and training or validation accuracy of developed model is plotted. Based on the validation plots, the VGG16 model exhibits the best accuracy in both the training and validation phases. The achieved training and validation accuracies are influenced by many aspects, such as the quality and amount of the data, the particular implementation approach, and the chosen training hyperparameters (e.g., learning rate, optimizer, etc.). In this research, 80 epochs were used for the training and validation procedures. The optimizer used in this research was adaptive moment estimation (Adam), with a learning rate 0.01. In a recent research conducted by Abubeker K M et al. [20] a new deep learning framework for medical picture classification, named B2Net, was addressed. The researchers claimed that using data augmentation and regularization approaches resulted in enhanced accuracy. Techniques such as learning rate schedules, dropout, and early stopping have optimized the model's performance.

However, training accuracy alone is not a sufficient metric to judge the model's performance, as it may overfit the training data if the number of epochs is too high. Validation accuracy measures how well the model generalizes to unseen data. Initially, as the model trains, both the training and validation accuracies tend to increase. The validation accuracy typically tracks the training accuracy closely, as the model learns valuable features from the training data. However, the validation accuracy may plateau or even decrease after a certain point. This is a sign that the model is starting to overfit the training data, meaning it's becoming too specialized in capturing noise or irrelevant details from the training set that don't generalize well to new data. This research shows that after the 30th epoch, the trading accuracy is almost constant but still increases. After the 50th epoch, there is a slight variation in training and validation accuracy, indicating the model has mature and we stop the training process at the 80th epoch.

D. Performance evaluation of the model

In practical applications, assessing image captioning models extends beyond the direct use of accuracy, sensitivity, specificity, and F1 score. Using specialized metrics like BLEU, METEOR, ROUGE, CIDEr, and SPICE is standard practice for evaluating a model's performance in picture captioning. The metrics consider the presence of specific words and the degree of overlap between n-grams, the semantic content, and the general coherence of the generated captions about the reference captions.

TABLE III. TEST RESULTS OF THE DEVELOPED VGG16-LSTM FRAMEWORK DEPLOYED IN RASPBERRY PI 4B SINGLE BOARD COMPUTER.

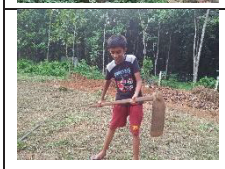
Input Image	Original Caption	Caption generated by VGG16-LSTM Model
	1. An agriculture land with coconut trees and vegetables. 2. A green land showing coconut trees and plants. 3. An agriculture land with coconut trees and some plants. 4. A land with coconut trees, a home and plants. 5. An agriculture land with coconut trees , a home and plants.	An agricultural land with coconut trees and some plants.
	1. A lion is sleeping in a green filed in front of a fountain. 2. A lion sleeping in front of a fountain with green background 3. A lion in front of a fountain with green background. 4. A lion is sitting in a green lone in front of a rock. 5. A lion sleeping in a green lone in front of a rock.	A lion is sleeping in a green filed in front of a fountain.
	1. A boy is playing in a dirty filed with hoe. 2. A boy with blue shirt and red half pant with hoe. 3. A boy with blue shirt and red nikar with hoe on a filed. 4. A boy with blue shirt is playing in a dirty land with hoe. 5. A boy with blue shirt and red half pant with a tool.	A boy with blue shirt and red nikar with hoe on a filed.
	1. A crowd of people working in sea side. 2. A large number of people looking in fish net. 3. A crowd of people with fish net on sea side. 4. A crowed with fish net on sea side. 5. A group of fisher man in a sea side with fish net.	A crowd of people with fish net on sea side.

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TABLE IV. BLEU-ROUGUE PERFORMANCE MATRIX OF THE DEVELOPED VGGNET-16 MODEL.

SI No.	Reference Image	BLEU score	Brevity Penalty	Length Ratio	Translation Length	Reference Length	Rouge1	Rouge2	RougeL	Rouge Lsum
1		0.553	0.978	0.978	45	46	0.717	0.451	0.717	0.716
2		0.688	1.000	1.10	65	59	0.799	0.642	0.762	0.762
3		0.656	1.000	1.04	65	62	0.750	0.588	0.707	0.707
4		0.373	0.860	0.869	40	46	0.565	0.310	0.561	0.558

In table III, the real-time image captioning using VGG16-LSTM network is presented. The research outcomes observed that the model is showing highest accuracy and efficacy throughout the testing process. Due to the nature of producing word sequences in picture captioning, conventional classification metrics may need to be more readily relevant. The Bilingual Evaluation Understudy (BLEU) metric is used to assess the degree of similarity between generated captions and reference captions. The recall-oriented understudy for gisting evaluation (ROUGE) metric quantifies the degree of similarity in terms of n-gram overlap between the generated captions and the reference captions. The assessment of the generated captions included ROUGE scores, namely ROUGE-N for n-gram overlap and ROUGE-L for longest common subsequence. Additionally, human review was conducted to evaluate generated captions' quality, relevance, and coherence might provide significant insights.

The brevity penalty has significance in metrics like BLEU, as it seeks to assess the effectiveness of machine-generated captions by considering their alignment with human perception regarding relevance, correctness, and completeness. The concept of the brevity penalty is used in the automated evaluation of BLEU to measure the efficacy of machine-generated translations or captions. The encouragement is for the generated captions to possess a more proximate length than the reference captions. According to the data shown in Table IV, The reference length for each generated caption is determined by BLEU, which is defined as the length of the closest reference caption in terms of the number of words. The phrase generated caption length or translation length refers to the numerical measurement of the length of a generated caption, quantified explicitly by the number of words it contains.

V. CONCLUSION

In conclusion, this work signifies notable advancements in the endeavour to create a digital environment that is more inclusive and user-friendly for those with visual impairments. The combination of computer vision and deep learning techniques, together with the integration of voice-based interaction, has shown considerable potential in mitigating the discrepancy between visual information and auditory perception experienced by those with visual impairments. The system has shown its capacity to generate exciting captions for images by using VGGNet-16 for image processing and LSTM for natural language processing. This amalgamation allows the framework to provide crucial insights into the visual content. The model in this research is trained using three datasets: Flickr8k, Flickr30k, and a custom dataset. Following this, the Raspberry Pi 4B single-board computer was used, which was outfitted with a graphics processing unit. The integration of a voice-based framework is consistent with the concept of providing information in a readily understandable and user-centric way. This facilitates individuals' independent access to visual content that was previously unavailable to them. The future trajectory of this research is centered on integrating other sensory modalities beyond visual and aural inputs, aiming to augment the overall user experience for those who are blind or visually impaired. Improving the framework's interoperability across various devices and platforms, such as smartphones, wearable devices, and smart glasses, would augment its accessibility and facilitate its seamless incorporation into users' daily activities. Expanding the linguistic functionalities of the framework to include more languages would augment its inclusivity, hence fostering a more diverse user base.

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